MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49WF POTABLE WATER TREATMENT PLANT

DESIGN OF TANK FDN 49WF-003,004 & 005

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
FDN	:	Foundation
RC	:	Reinforced Concrete

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1. INTENT

This document is to prepare design of foundation for tanks **49WF-003,49WF-004 & 49WF-005**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah – Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from structural calculation & designed in excel. The stability of the foundation, the bearing pressure and RC details have been calculated in excel.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Input drawings attached in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 225 Kn/m^2 at 0.85 m below FGL & 500 Kn/m^2 at 1.5m below FGL (Refer Table 10 - BH-21 of Geotechnical Report: MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

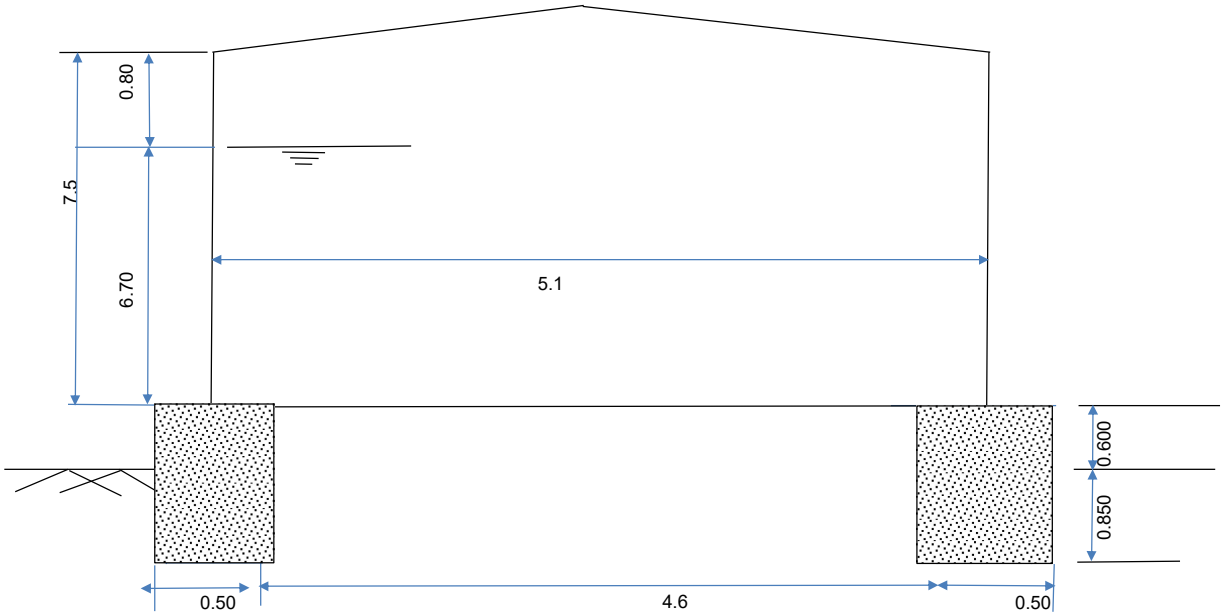
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Tank Loading Data of 49WF-003

	Loading Type		
Weight, V	Empty (W)	214	kN
	Operating (Wo)	1757	kN
	Test (Wt)	1823	kN

Summary of Tank Specification



Diameter of Tank, D	=	5.1 m	
Height of Tank, H	=	7.5 m	
Height of Tank Fluid, H_L	=	6.7 m	
Inner Diameter of Concrete Ring Wall, ID	=	4.6 m	
Outer Diameter of Concrete Ring Wall, OD	=	5.6 m	
Height of Concrete Ring Wall, h	=	1.450 m	
ThickNess of Concrete Ring Wall, t	=	0.5 m	
Top of Asphalt Board	=	0.6 m	(Above Finished Level)
Unit Weight of Liquid, g_L	=	10 kN/m3	
Unit Weight of Soil, g_s	=	20 kN/m3	
Unit Weight of Concrete, g_c	=	24 kN/m3	

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Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	31.00	245.00
	Operating	31.00	245.00
Seismic Loads	Operating	42.00	154.00

Foundation Weight

a) Ring Wall Weight = 278.78 kN

a) Inside Soil Weight = 481.95 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 8.01 m^2

Full cross-sectional area for operating condition, Ao= 24.63 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 9.39 m^3

Sectional modulus for operating condition,Zo = 17.24 m^3

Check of Minimum Wall ThickNess (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickNess of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = g_L{*}H_L + g_s * h$

Soil pressure under the ringwall, $P = w / t + 0.5\, g_L\, H_L + gc\, h$

Solving the above 2 equations, $g_L{*}H_L + g_s * h = w / t + 0.5\, g_L\, H_L + gc\, h$

or, $t = w /(0.5 g_L{*}H_L + h(g_s - gc))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, (W/pD) = 13.36 kN/m

So, Required Min Wall THK.= 0.482 m = 482 mm < 500 mm OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	1757.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	128.40	31.00	245.00
	Operating	LC103	D + F1 + W	1757.00	31.00	245.00
		LC104	D + F1+ 0.7 E	1757.00	29.40	107.80
		LC105	0.6 (D + F1) + 0.7 E	1054.20	29.40	107.80
	Test	LC106	D + F2	1823.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 225 kN/m2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 299.25 kN/m2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m2	Qmax= V/A - M/Z kN/m2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	2517.74	0.00	102.22	102.22	2.20	OK
LC102	407.18	289.95	81.70	19.95	3.66	OK
LC103	2517.74	289.95	119.04	85.40	2.51	OK
LC104	2517.74	150.43	110.95	93.50	2.70	OK
LC105	1814.94	150.43	82.41	64.96	3.63	OK
LC106	2583.74	0.00	104.90	104.90	2.14	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	Hresist (kN)	Hacting (kN)	Fs	Sliding Check
	V (kN)					
LC101	2517.74	0.4	1007.09	0.00	-	-
LC102	407.18	0.4	162.87	31.00	5.3	OK
LC103	2517.74	0.4	1007.09	31.00	32.5	OK
LC104	2517.74	0.4	1007.09	29.40	34.3	OK
LC105	1814.94	0.4	725.97	29.40	24.7	OK
LC106	2583.74	0.4	1033.49	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	Mresist (kNm)	Macting (kNm)	Fs	Overturning Check
	V (kN)					
LC101	2517.74	2.8	7049.66	0.00	-	-
LC102	407.18	2.8	1140.12	289.95	3.9	OK
LC103	2517.74	2.8	7049.66	289.95	24.3	OK
LC104	2517.74	2.8	7049.66	150.43	46.9	OK
LC105	1814.94	2.8	5081.82	150.43	33.8	OK
LC106	2583.74	2.8	7234.46	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 , Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	2459.80	0.00	20.43	13.02
Short term	Empty	LC102	0.9 D + 1.6W	192.60	392.00	20.43	13.02
	Operating	LC103	1.2 (D + F1) +1.6 W	2108.40	392.00	20.43	13.02
		LC104	1.2 (D + F1) + E	2108.40	154.00	20.43	13.02
		LC105	0.9 (D + F1) + E	1581.30	154.00	20.43	13.02
	Test	LC106	1.4 (D + F2)	2552.20	0.00	20.43	13.02

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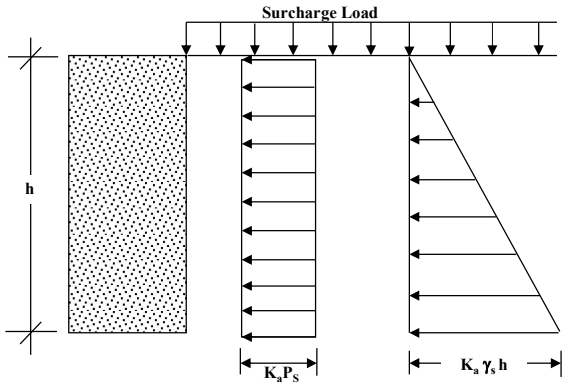
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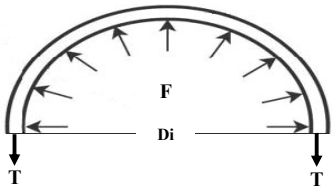
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Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m2
LC101	2459.80	0.00	120.41
LC102	192.60	392.00	39.53
LC103	2108.40	392.00	133.31
LC104	2108.40	154.00	115.04
LC105	1581.30	154.00	89.23
LC106	2552.20	0.00	124.94

Maximum Surcharge Load, Ps = 133.31 kN/m2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 66.66 kN/m2

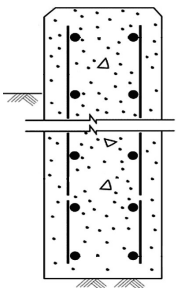
P2 = $1.4 * g_S * h * K_a$ = 20.30 kN/m2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 111.3679 kN/m

Hoop Tension Load

T= 1/2 * F * ID = 256.1461 kN



Hoop Tension Bar

1. Horizontal reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 1812.5 mm^2

Minimum As = 0.0025 x b x D = 1812.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D12= 2041.4069 mm^2 OK

2. Vertical reinforcement

Minimum As = 0.0015 * b * D = 750 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 6 ea - D10 (@ 167mm)

=942mm2 > Required As OK

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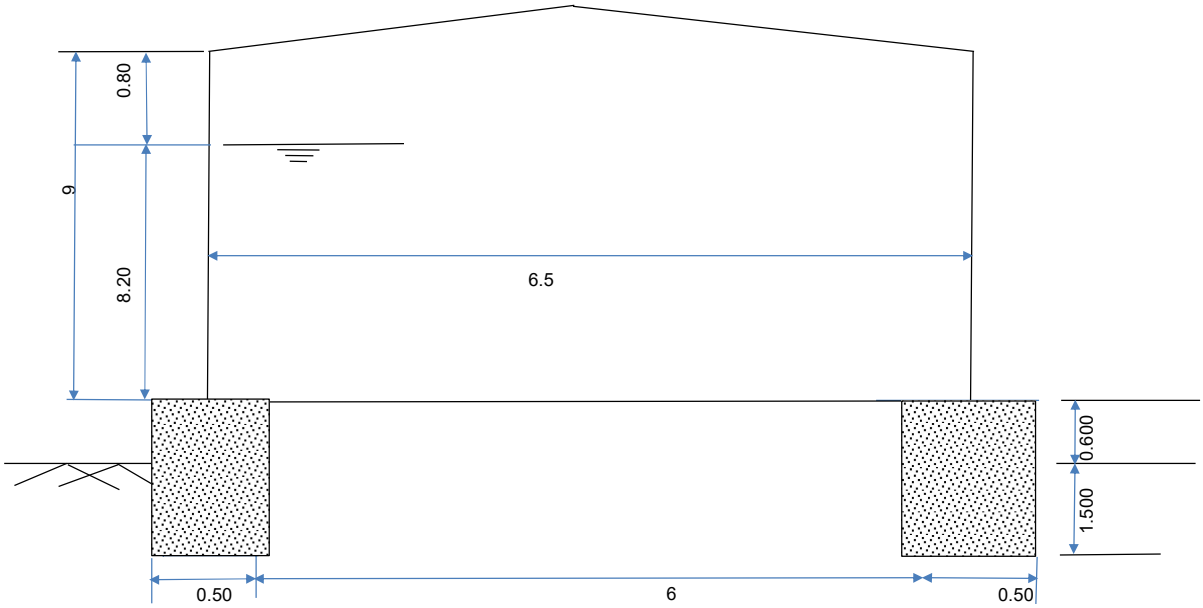
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Tank Loading Data 49WF-004 & 005

	Loading Type		
Weight, V	Empty (W)	214	kN
	Operating (Wo)	3083	kN
	Test (Wt)	3250	kN

Summary of Tank Specification



Diameter of Tank, D	=	6.5 m	
Height of Tank, H	=	9 m	
Height of Tank Fluid, H_L	=	8.2 m	
Inner Diameter of Concrete Ring Wall, ID	=	6 m	
Outer Diameter of Concrete Ring Wall, OD	=	7 m	
Height of Concrete Ring Wall, h	=	2.100 m	
ThickNess of Concrete Ring Wall, t	=	0.5 m	
Top of Asphalt Board	=	0.6 m	(Above Finished Level)
Unit Weight of Liquid, g_L	=	10 kN/m3	
Unit Weight of Soil, g_S	=	20 kN/m3	
Unit Weight of Concrete, g_C	=	24 kN/m3	

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Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	31.00	245.00
	Operating	31.00	245.00
Seismic Loads	Operating	42.00	154.00

Foundation Weight

a) Ring Wall Weight = 514.59 kN

a) Inside Soil Weight = 1187.52 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 10.21 m^2

Full cross-sectional area for operating condition, Ao= 38.48 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 15.50 m^3

Sectional modulus for operating condition,Zo = 33.67 m^3

Check of Minimum Wall ThickNess (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickNess of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + 0.5\, \gamma_L H_L + \gamma_c h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + 0.5\, \gamma_L H_L + \gamma_c h$

or, $t = w / (0.5\gamma_L * H_L + h(\gamma_s - \gamma_c))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, (W/pD) = 10.48 kN/m

So, Required Min Wall THK.= 0.321 m = 321 mm < 500 mm OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	3083.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	128.40	31.00	245.00
	Operating	LC103	D + F1 + W	3083.00	31.00	245.00
		LC104	D + F1+ 0.7 E	3083.00	29.40	107.80
		LC105	0.6 (D + F1) + 0.7 E	1849.80	29.40	107.80
	Test	LC106	D + F2	3250.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 500 kN/m2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 665.00 kN/m2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m2	Qmax= V/A - M/Z kN/m2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	4785.11	0.00	124.34	124.34	4.02	OK
LC102	642.99	310.10	82.99	42.97	8.01	OK
LC103	4785.11	310.10	133.55	115.13	4.98	OK
LC104	4785.11	169.54	129.37	119.30	5.14	OK
LC105	3551.91	169.54	97.33	87.26	6.83	OK
LC106	4952.11	0.00	128.68	128.68	3.89	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	Hresist (kN)	Hacting (kN)	Fs	Sliding Check
	V (kN)					
LC101	4785.11	0.4	1914.05	0.00	-	-
LC102	642.99	0.4	257.20	31.00	8.3	OK
LC103	4785.11	0.4	1914.05	31.00	61.7	OK
LC104	4785.11	0.4	1914.05	29.40	65.1	OK
LC105	3551.91	0.4	1420.77	29.40	48.3	OK
LC106	4952.11	0.4	1980.85	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	Mresist (kNm)	Macting (kNm)	Fs	Overturning Check
	V (kN)					
LC101	4785.11	3.5	16747.90	0.00	-	-
LC102	642.99	3.5	2250.48	310.10	7.3	OK
LC103	4785.11	3.5	16747.90	310.10	54.0	OK
LC104	4785.11	3.5	16747.90	169.54	98.8	OK
LC105	3551.91	3.5	12431.70	169.54	73.3	OK
LC106	4952.11	3.5	17332.40	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 , Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	4316.20	0.00	33.18	26.96
Short term	Empty	LC102	0.9 D + 1.6W	192.60	392.00	33.18	26.96
	Operating	LC103	1.2 (D + F1) +1.6 W	3699.60	392.00	33.18	26.96
		LC104	1.2 (D + F1) + E	3699.60	154.00	33.18	26.96
		LC105	0.9 (D + F1) + E	2774.70	154.00	33.18	26.96
	Test	LC106	1.4 (D + F2)	4550.00	0.00	33.18	26.96

PROPRIETARY & CONFIDENTIAL

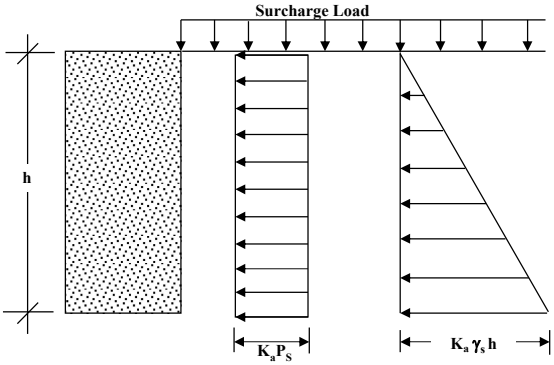
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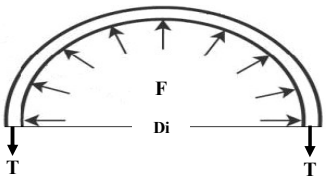
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Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m2
LC101	4316.20	0.00	130.07
LC102	192.60	392.00	20.34
LC103	3699.60	392.00	126.03
LC104	3699.60	154.00	117.20
LC105	2774.70	154.00	89.33
LC106	4550.00	0.00	137.12

Maximum Surcharge Load, Ps = 137.12 kN/m2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 68.56 kN/m2

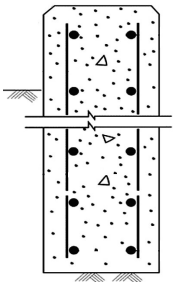
P2 = $1.4 * \gamma_S * h * K_a$ = 29.40 kN/m2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 174.844 kN/m

Hoop Tension Load

T= 1/2 * F * ID = 524.532 kN



Hoop Tension Bar

1. Horizontal reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 2625 mm^2

Minimum As = 0.0025 x b x D = 2625 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 28 Nos. D12= 2857.9697 mm^2 OK

2. Vertical reinforcement

Minimum As = 0.0015 * b * D = 750 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 6 ea - D10 (@ 167mm)

=942mm2 > Required As OK

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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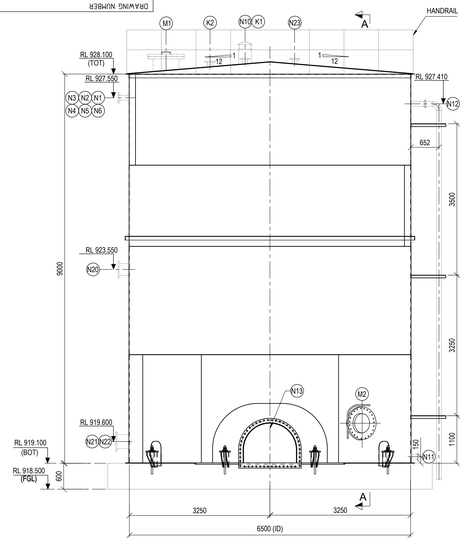
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ATTACHMENT 1 INPUT DRAWINGS

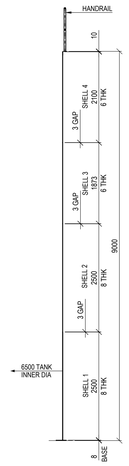
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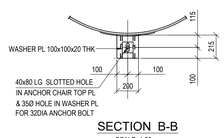
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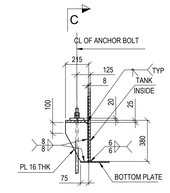
TANK ELEVATION
(SCALE 1:50)



SECTION A - A
(SCALE - 1: 50)



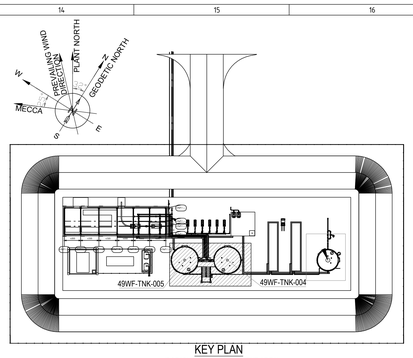
SECTION B-B
SCALE- 1:20



TYP DETAIL OF ANCHOR CHAIR
SCALE- 1:20

MATERIAL SPECIFICATION	
SHELL, BOTTOM, ANNULAR, ROOF	EN 10025 GR.S275 J0
COD	EN 10025 GR.S275 J0
MANWAY NECKS/ MANWAY FLANGES	EN 10025 GR.S275 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A53 Gr.B/ ASTM A106 Gr.B
DELETED	DELETED
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	NON METALLIC FLAT RING GASKET 1.5 mm THICK PTFE, CNAF
GASKETS FOR MANWAY/ COD	3mm THK. COMPRESSED NON-ASBESTOS SHEETS.
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	EN 10025 GR.S275 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	EN 10025 GR.S275 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 TYPE E
PIPE FITTINGS	SA234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	EN 10025 GR.S275 J0
EARTHING LUGS	EN 10025 GR.S275 J0
ANCHOR BOLTS	ASTM F1554 GR.36
CURB ANGLE	EN 10025 GR.S275 JR
NAME PLATE	STAINLESS STEEL(SS 316)

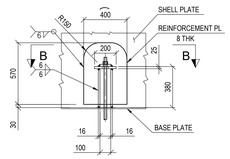
FOUNDATION LOAD DATA:					
BASIC LOAD CASE	VERTICAL COMPRESSION	VERTICAL UPLIFT	HORIZONTAL SHEAR	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT)	SLAB MOMENT
	(KN)	(KN)	(KN)	(KN-M)	(KN-M)
DEAD LOAD (NOMINAL THICKNESS)	214	–	–	–	–
DEAD LOAD (CORRODED THICKNESS)	177	–	–	–	–
ROOF PIPE LIQUID LOAD	53	–	–	–	–
SHELL PIPE LIQUID LOAD	53	–	–	–	–
LIVE LOAD	94	–	–	–	–
TANK LIQUID LOAD (OPERATING)	2669	–	–	–	–
TANK LIQUID LOAD (TEST)	2930	–	–	–	–
WIND LOAD ON SHELL	–	–	31	142	–
WIND LOAD ON ROOF	–	32	–	103	–
SEISMIC LOAD	95	95	42	154	192



KEY PLAN
(POTABLE WATER TREATMENT AREA)
(SCALE-1:300)

NOZZLE SCHEDULE (FOR TANK NO 49WF-TNK-004)												
NOZ.	SERVICE	SIZE (DN)	SCH	RATING #	TYPE	ELEV "H"	PROJ "P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA *	RF PAD THK *	REMARK
N1	POTABLE WATER INLET	50	Sch XS	CL 150	RFSO	8.450	175	-	160°	SHELL	200	6	
N2	FIRE WATER RECIRCULATION LINE	100	Sch XS	CL 150	RFSO	8.450	200	-	305°	SHELL	305	6	
N3	POTABLE WATER RECIRCULATION LINE	50	Sch XS	CL 150	RFSO	8.450	175	-	315°	SHELL	200	6	
N4	TANKER FILLING	50	Sch XS	CL 150	RFSO	8.450	175	-	190°	SHELL	200	6	
N5	POTABLE WATER RECIRCULATION LINE	100	Sch XS	CL 150	RFSO	8.450	175	-	325°	SHELL	305	6	
N6	FIRE WATER RECIRCULATION LINE FROM PUMP 39LA-PMP-013	200	Sch XS	CL 150	RFSO	8.450	200	-	295°	SHELL	485	6	
N10	TANK VENT	100x100	-	COVER ARRANGEMENT		-	400	1500	112.5°	ROOF	TBC	TBC	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300	-	50°	SHELL	305	10	
N12	OVERFLOW	100	Sch XS	CL 150	RFSO	8.310	200	-	45°	SHELL	305	6	
N13	CLEAN OUT DOOR (COD)-hxb	900x1200	-	API 650		-	-	-	90°	SHELL	1372x3025	10	
N20	POTABLE WATER OUTLET	350	Sch XS	CL 150	RFSO	4.450	250	-	335°	SHELL	750	10	
N21	FIRE WATER OUTLET	200	Sch XS	CL 150	RFSO	0.500	250	-	0°	SHELL	485 (BOTH SIDE)	10	INTERNAL 20 EXTENSION
N22	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	0.500	200	-	15°	SHELL	305	10	
N23	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	100	1500	292.5°	ROOF	305	6	
K1	LEVEL TRANSMITTER LT 97527	100	Sch XS	CL 150	RFSO	-	100	1500	247.5°	ROOF	305	6	
K2	LEVEL TRANSMITTER LT 92010	100	Sch XS	CL 150	RFSO	-	100	1500	157.5°	ROOF	305	6	
M1	MANHOLE C/W HINGED COVER	600	-			-	500	2600	202.5°	ROOF	1150	6	
M2	MANHOLE C/W DAVIT	750	-	API 650		0.925	175	-	180°	SHELL	1545	10	

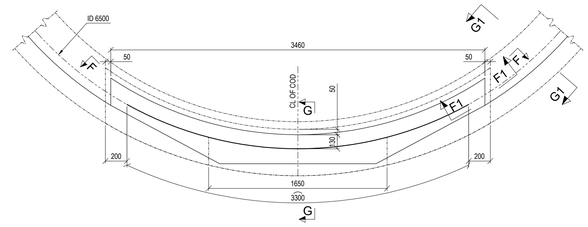
* REFER NOTE NO. 12 & 17



SECTION C-C
SCALE – 1: 20

NOZZLE SCHEDULE (FOR TANK NO 49WF-TNK-005)												
NOZ.	SERVICE	SIZE (DN)	SCH	RATING #	TYPE	ELEV "H"	PROJ "P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA *	RF PAD THK *	REMARK
N1	POTABLE WATER INLET	50	Sch XS	CL 150	RFSO	8.450	175	-	170°	SHELL	200	6	
N2	FIRE WATER RECIRCULATION LINE	100	Sch XS	CL 150	RFSO	8.450	200	-	55°	SHELL	305	6	
N3	POTABLE WATER RECIRCULATION LINE	50	Sch XS	CL 150	RFSO	8.450	175	-	45°	SHELL	200	6	
N4	TANKER FILLING	50	Sch XS	CL 150	RFSO	8.450	175	-	160°	SHELL	200	6	
N5	POTABLE WATER RECIRCULATION LINE	100	Sch XS	CL 150	RFSO	8.450	175	-	35°	SHELL	305	6	
N6	FIRE WATER RECIRCULATION LINE FROM PUMP 39LA-PMP-013	200	Sch XS	CL 150	RFSO	8.450	200	-	65°	SHELL	485	6	
N10	TANK VENT	100x100	-	COVER ARRANGEMENT		-	400	1500	247.5°	ROOF	TBC	TBC	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300	-	310°	SHELL	305	10	
N12	OVERFLOW	100	Sch XS	CL 150	RFSO	8.310	200	-	315°	SHELL	305	6	
N13	CLEAN OUT DOOR (COD)-hxb	900x1200	-	API 650		-	-	-	270°	SHELL	1372x3025	10	
N20	POTABLE WATER OUTLET	350	Sch XS	CL 150	RFSO	4.450	250	-	25°	SHELL	750	10	
N21	FIRE WATER OUTLET	200	Sch XS	CL 150	RFSO	0.500	250	-	0°	SHELL	485 (BOTH SIDE)	10	INTERNAL 20 EXTENSION
N22	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	0.500	200	-	345°	SHELL	305	10	
N23	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	100	1500	67.5°	ROOF	305	6	
K1	LEVEL TRANSMITTER LT 97529	100	Sch XS	CL 150	RFSO	-	100	1500	112.5°	ROOF	305	6	
K2	LEVEL TRANSMITTER LT 92011	100	Sch XS	CL 150	RFSO	-	100	1500	202.5°	ROOF	305	6	
M1	MANHOLE C/W HINGED COVER	600	-			-	500	2600	157.5°	ROOF	1150	6	
M2	MANHOLE C/W DAVIT	750	-	API 650		0.925	175	-	180°	SHELL	1545	10	

REFER NOTE NO. 12 & 17



DETAIL OF CLEAN OUT DOOR (COD)
(SIZE: 900x1200)
SCALE- 1:20

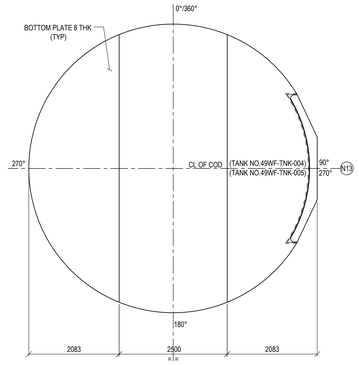
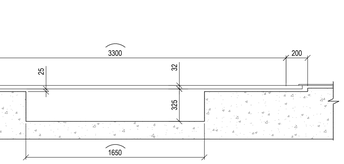
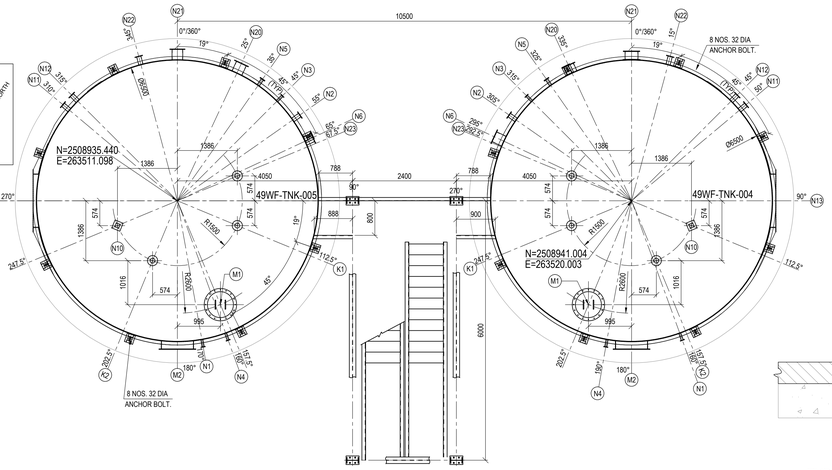


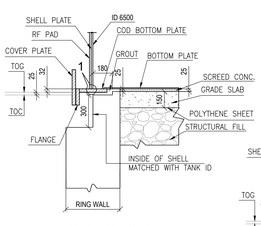
PLATE ARRANGEMENT OF BOTTOM PLATE
(SCALE - 1: 50)



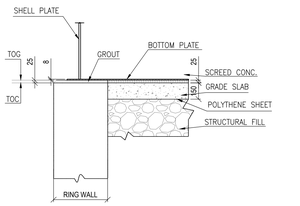
SECTION F-F



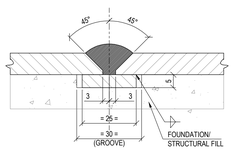
PALN
(SCALE - 1: 50)



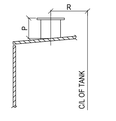
SECTION G-G



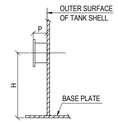
SECTION G1-G1



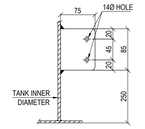
TYPICAL BOTTOM PLATE JOINT DETAIL
(SCALE - 1: 10)



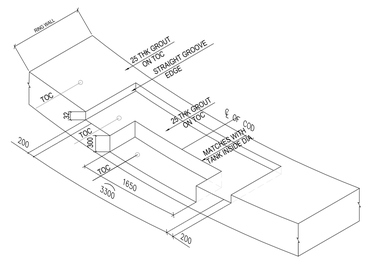
ROOF NOZZLES / FITTINGS



SHELL NOZZLES / FITTINGS



DETAIL OF EARTHING LUGS



3D VIEW OF FOUNDATION NOTCH (FOR COD)

LEGEND:

ID - INNER DIAMETER
OD - OUTER DIAMETER
COD - CLEAN OUT DOOR
LG - LONG
FGL - FINISHED GROUND LEVEL.
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
RL - REDUCED LEVEL
RF - REINFORCEMENT
TOC - TOP OF CONCRETE
GST - GUSSET
PL - PLATE
STIFF - STIFENER
THK - THICKNESS
CLT - CLEAT
UNO - UNLESS NOTED OTHERWISE
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
NOZ - NOZZLE
DN - DIAMETER NOMINAL
SCH - SCHEDULE
ELVN - ELEVATION
PROJ - PROJECTION
MM - MILLIMETER
TYP - TYPICAL
GA - GENERAL ARRANGEMENT
DRG - DRAWING
CL - CENTER LINE
TOS - TOP OF STEEL
WP - WORKING POINT
N/S - NEAR SIDE
F/S - FAR SIDE
DIA - DIAMETER
C/F - CUT FORM
TBC - TO BE CONFIRMED
TOG - TOP OF GRATING
N/A - NOT APPLICABLE
ATM - ATMOSPHERIC
MIN - MINIMUM
TOP - TOP OF PAVING

DESIGN DATA			
TANK DETAILS	LOCATION		POTABLE WATER TREATMENT AREA
	TANK NAME		POTABLE WATER STORAGE TANK
	EQUIPMENT TAG No.		49WF-TNK-004 & 49WF-TNK-005
	TANK DIMENSIONS		INSIDE DIAMETER : 6500mm, HEIGHT: 9000mm
	QUANTITY (NUMBER)		2
	ROOF TYPE & SLOPE		SLOPED CONE ROOF (SLOPE 1 IN 12) CONICAL APEX UP
	BOTTOM TYPE & SLOPE		SLOPED BOTTOM (SLOPE 1 IN 120) CONICAL APEX UP
	FIELD ERECTED/SHOP FABRICATED		FIELD ERECTED
	STORED FLUID		POTABLE WATER
DESIGN CODE	DESIGN CODE		API 650-TWELFTH EDITION, MARCH 2013, ADDENDUM 3, AUGUST 2018
OPERATING DATA	OPERATING TEMPERATURE		40° C
	NET WORKING CAPACITY		245 m^3
	MAXIMUM CAPACITY		274 m^3
	OPERATING LEVEL	MINIMUM	870 mm
		MAXIMUM	8250 mm
	OPERATING DENSITY OF CONTENT		1.0 t/m^3
MECHANICAL & PROCESS DATA	DESIGN TEMPERATURE		50° C
	DESIGN PRESSURE (INTERNAL/EXTERNAL)		ATM/ATM
	TEST PRESSURE		ATM
	HYDRO STATIC TEST PRESSURE		FULL OF WATER
	DESIGN DENSITY OF CONTENT		1.0 t/m^3
	TESTING		AS PER API 650
	INSPECTION		AS PER API 650
	DENSITY OF SCALE		N/A
	DESIGN SCALE THICKNESS	TANK SHELL/ROOF	N/A
	DESIGN FILL RATE		5 m/hr
	DESIGN EMPTY RATE		232 m3/hr
	CORROSION ALLOWANCE	SHELL	1.6 mm
		ROOF PLATE	0 mm
		BOTTOM PLATE / ANNULAR PLATE	1.6 mm
		ANCHOR BOLT	0 mm
		STRUCTURAL	1.6 mm
WIND LOAD DATA	BASIC WIND SPEED		43 m/sec. (155 Km/hr.)
	SITE EXPOSURE		C
	VELOCITY PRESSURE EXPOSURE COEFFICIENT		K_z= AS PER ELEVATION
	TOPOGRAPHIC FACTOR		K_{zt}=1.0
	WIND DIRECTIONALITY FACTOR		K_d=0.95
SEISMIC LOAD DATA	SITE CLASS		B
	SEISMIC USER GROUP		SUG - I
	IMPORTANCE FACTOR	(I)	1
	SCALING FACTOR	(Q)	2/3
	MAXIMUM DESIGN LIQUID LEVEL	(H)	8.250 m
	CO-EFFICIENT TO ADJUST THE SPECTRAL ACCELERATION FROM 5% TO 0.5% DAMPING	(K)	1.50
	SEISMIC CO-EFFICIENTS		S_s= 0.1g, S_1= 0.02g, F_a = 1.0, F_v= 1.0 S_{ds} = 0.067g, S_{d1} = 0.013g, T_s = 4 sec.
OTHER DATA	ROOF LIFE LOAD		1.2 kPa
	FOUNDATION TYPE (RING WALL/SLAB)		RING WALL
	PWHT REQUIRED(YES/NO)- IF YES,SPECIFY REASON FOR PWHT		NO
	REASON FOR PWHT/PROCESS REASON/PER API CODE)		AS PER API 650
	RADIOGRAPHY		AS PER CL 8.1 - API 650
	AGITATOR RAKE REQUIRED	(YES/NO)	NO
		EQUIPMENT TYPE (AGITATOR/RAKE)	N/A
		AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)	N/A
	INSULATION	SHELL	N/A
		ROOF	N/A
	LINER REQUIRED	(YES/NO)	NO
		LINER MATERIAL	N/A
		LINER THICKNESS (ROOF/SHELL/BOT)	N/A
	PAINTING	INTERNAL	YES
		EXTERNAL	YES
	PIPE SUPPORT		YES
	COMMON STAIRWAY		YES
	ROOF CIRCUMFERENTIAL HANDRAIL		YES
	WIND GIRDER		AS REQUIRED
	CATHODIC PROTECTION		NO
	LEAK DETECTION (APPURTENANCE)		NO
	INTERNAL BAFFLES		NO
	EARTHING LUGS		YES

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. BOLT HOLES IN NOZZLE FLANGES & MANHOLES SHALL STRADDLE VERTICAL CENTER LINES ON SHELL & NORTH SOUTH CENTERLINES ON ROOF.
5. NOZZLE PROJECTION ON SHELL SIDE REFERS FROM EXTERNAL SURFACE OF SHELL PLATE TO FACE OF FLANGE. NOZZLE PROJECTION ON ROOF SIDE REFERS FROM TOP OF ROOF PLATE TO FACE OF FLANGE.
6. VERTICAL SHELL JOINTS SHOULD PREFERABLY BE OFFSET BY 1000 MM.
7. CURB ANGLE JOINTS SHALL STAGGER ATLEAST 300 MM WITH VERTICAL SHELL PLATE WELD.
8. REINFORCEMENT PADS TO HAVE TELLTALE HOLES OF 6 MM DIA AND SHALL BE TESTED AT 100 kPa PRESSURE WITH SOAP SOLUTION AND THEN FILLED WITH HARD GREASE.
9. FLANGE (COD & MANHOLE) GASKET SEATING SURFACE SHALL BE PROVIDED WITH 0.0032 TO 0.0063 mm (3.2 TO 6.3 µm) ROUGHNESS FINISH.
10. EARTHING LUG LOCATION SHALL BE DONE SUIT TO SITE CONDITION.
11. COD GROOVE DIMENSION ARE FINALIZED BASED ON COD BOTTOM PLATE DIMENSION.
12. STIFFENERS FOR NOZZLE, IF REQUIRED SHALL BE INSTALLED AT SUIT TO SITE.
13. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 49WF-EG -ME- GAD - 2002, 2003 & 2004.
14. PAINTING SHALL CONFORM TO MAADEN SPECIFICATION NUMBER MD-101-SM-EG-ME-SPC-0007
15. ALL NOZZLE FLANGES SHALL BE IN ACCORDANCE WITH ASME B16.5 FOR SIZE UPTO DN600 AND TO ASME B16.47 SERIES A FOR DN750 AND ABOVE.
16. FOR HANDRAIL AND STAIR RELATED INFORMATION REFER DRG. NO. MD-216-1000-EG-ST-DED-1002 & 1003 (LATEST).
17. DN50 PIPE SUPPORT TO BE PROVIDED AT SUIT TO SITE CONDITION.

THIS DRAWING SUPERSEDES MAADEN DRAWING NUMBER MD-216-49WF-EG-ST-GAD-1001-C01 DATE: 07/03/2020

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	D01	30/10/2023	AS BUILT	SDR	PRI	KDU	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	C02	03/09/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C02	SDR	PRI	KDU	AG	
MD-216-49WF-EG-ME-GAD-1008	POTABLE WATER STORAGE TANK 49WF-TNK-004	C01	07/03/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C01	SJH	PRI	KDU	AG	
MD-216-49WF-EG-ME-GAD-1009	POTABLE WATER STORAGE TANK 49WF-TNK-005	A01	31/01/2020	ISSUED FOR REFERENCE	SJH	PRI	KDU	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_S_G_26210_001 REVISION: 3
OUTOTEC DRAWING NUMBER: 900707__GAU01__AG32__MTL01__53059 DOCUMENT ID: OU100671395 REVISION: 3

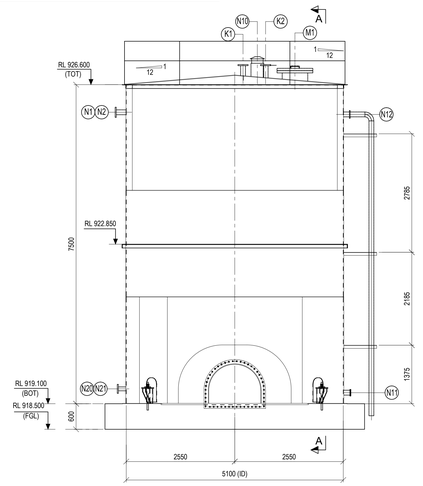
LARSEN & TOUBRO
This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

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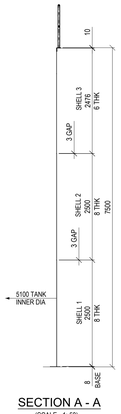
DRAWN	SJH	28/01/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	PRI	30/01/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	KDU	29/01/2020	POTABLE WATER SUPPLY, TRT & DIST
APPROVED	AG	31/01/2020	PLAN, ELVN, DETAIL OF TANK 49WF-TNK-004,005
EMPLOYER			
SCALE	1:100,20	A0	DRAWING NUMBER: MD - 216 - 49WF- EG - ME - GAD - 2001 REV: D01

DRAWING NUMBER: MD - 216 - 49WF-EG - ME - GAD - 2005 | REV: D01



TANK ELEVATION

(SCALE 1:50)



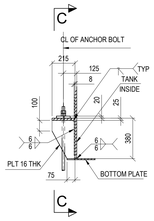
SECTION A - A

(SCALE - 1: 50)



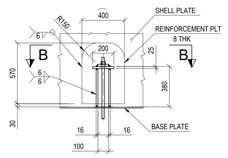
SECTION B-B

SCALE- 1:20



TYP DETAIL OF ANCHOR CHAIR

SCALE- 1:20



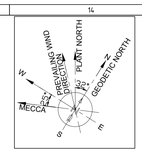
SECTION C-C

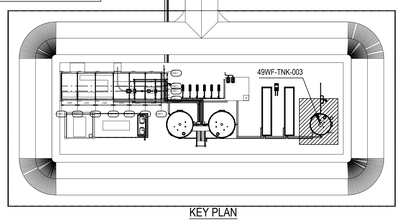
SCALE– 1: 20

MATERIAL SPECIFICATION	
SHELL, BOTTOM, ANNULAR, ROOF	EN 10025 GR.S275 J0
COD, MANHOLE	EN 10025 GR.S275 J0
MANWAY NECKS/ MANWAY FLANGES	EN 10025 GR.S275 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A106 Gr.B
DELETED	DELETED
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	NON METALLIC FLAT RING GASKET 1.5 mm THICK
GASKETS FOR MANWAY/ COD	3mm THK. COMPRESSED NON-ASBESTOS SHEETS.
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	EN 10025 GR.S275 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	EN 10025 GR.S275 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 TYPE E, GRADE B
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	EN 10025 GR.S275 J0
EARTHING LUGS	EN 10025 GR.S275 J0
ANCHOR BOLTS	ASTM F1554 GR.36
CURB ANGLE	EN 10025 GR.S275 JR
NAME PLATE	STAINLESS STEEL(SS 316)

FOUNDATION LOAD DATA **					
BASIC LOAD CASE	VERTICAL COMPRESSION	VERTICAL UPLIFT	HORIZONTAL SHEAR	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT)	SLAB MOMENT
	(KN)	(KN)	(KN)	(KN-M)	(KN-M)
DEAD LOAD (NOMINAL THICKNESS)	214	–	–	–	–
DEAD LOAD (CORRODED THICKNESS)	177	–	–	–	–
ROOF PIPE LIQUID LOAD	53	–	–	–	–
SHELL PIPE LIQUID LOAD	53	–	–	–	–
LIVE LOAD	94	–	–	–	–
TANK LIQUID LOAD (OPERATING)	1343	–	–	–	–
TANK LIQUID LOAD (TEST)	1503	–	–	–	–
WIND LOAD ON SHELL	–	–	31	142	–
WIND LOAD ON ROOF	–	32	–	103	–
SEISMIC LOAD	95	95	42	154	192

** REFER NOTE 15





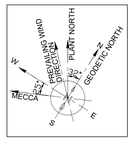
KEY PLAN

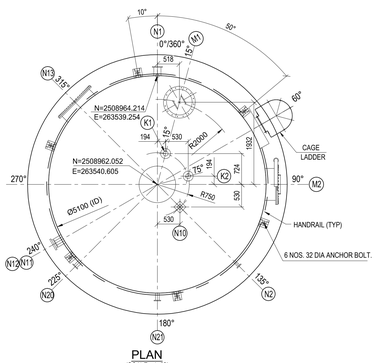
(POTABLE WATER TREATMENT AREA)

(SCALE-1:300)

NOZZLE TABLE												
NOZ	SERVICE	SIZE (DN)	SCH	RATING #	TYPE	ELEV "H"	PROJ "P"	RADIUS "R" *	ORIENTATION	LOCATION	RF PAD DIA *	RF PAD THK. *	REMARK
N1	RAW WATER INLET FROM 49WF-PMP-007 & 008	150	Sch XS	CL 150	RFSO	6.950	200	-	0°	SHELL	400	6	
N2	TANKER FILLING	50	Sch XS	CL 150	RFSO	6.950	175	-	135°	SHELL	-	-	
N10	TANK VENT	100x100 (TBC)	-	RECTANGULAR WITH COVER ARRANGEMENT		-	400	750	135°	ROOF	-	-	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.150	300	-	240°	SHELL	305	8	
N12	OVERFLOW	200	Sch XS	CL 150	RFSO	6.800	200	-	240°	SHELL	485	8	
N13	CLEAN OUT DOOR (COD)-hxb	900x1200	-	API 650		-	-	-	315°	SHELL	1372X2700	10	
N20	RAW WATER OUTLET	80	Sch XS	CL 150	RFSO	0.200	200	-	225°	SHELL	265	8	
N21	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	0.300	200	-	180°	SHELL	305	8	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	100	750	15°	ROOF	305	6	
K2	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	100	750	75°	ROOF	305	6	
M1	MANHOLE C/W HINGED COVER	600	-			-	500	2000	15°	ROOF	1150	6	
M2	MANHOLE C/W DAVIT	750	-	API 650		0.925	175	-	90°	SHELL	1545	10	

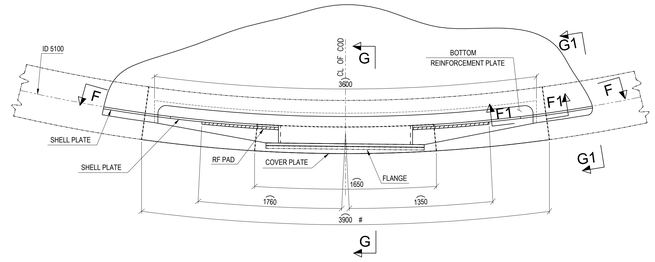
* REFER NOTE NO. 12





PLAN

(SCALE - 1: 50)

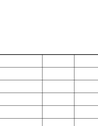


DETAIL OF CLEAN OUT DOOR (COD)

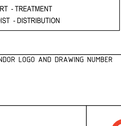
(# DIMENSION TO BE FINALISED BASED ON FOUNDATION RING WIDTH)

(COD SIZE 900x1200)

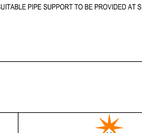
SCALE- 1:20



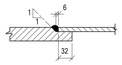
ROOF NOZZLES / FITTINGS



SHELL NOZZLES / FITTINGS



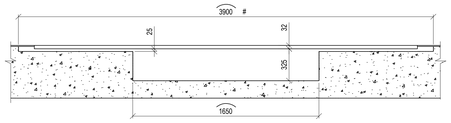
DETAIL OF EARTHING LUGS



SECTION F1-F1

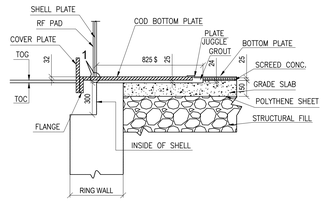


DETAIL-1



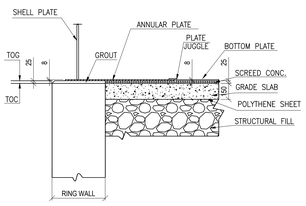
SECTION F-F

(# DIMENSION TO BE FINALISED BASED ON FOUNDATION RING WIDTH)



SECTION G-G

($ TO BE FINALISED BASED ON COD BOTTOM PLATE DIMENSION)



SECTION G1-G1

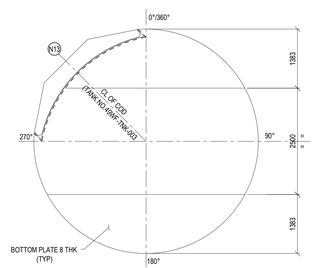
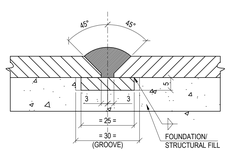


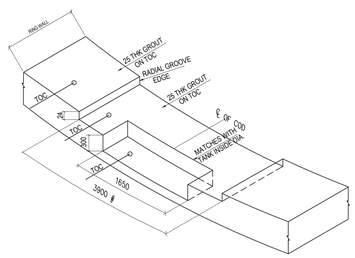
PLATE ARRANGEMENT OF BOTTOM PLATE

(SCALE - 1: 50)



TYPICAL BOTTOM PLATE JOINT DETAIL

(SCALE - 1: 10)



3D VIEW OF FOUNDATION NOTCH (FOR COD)

(# DIMENSION TO BE FINALISED BASED ON FOUNDATION RING WIDTH)

LEGEND:

ID - INNER DIAMETER
OD - OUTER DIAMETER
COD - CLEAN OUT DOOR
LG - LONG
FGL - FINISHED GROUND LEVEL
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
RL - REDUCED LEVEL
RF - REINFORCEMENT
TOC - TOP OF CONCRETE
GST - GUSSET
PLT - PLATE
STIFF - STIFENER
THK - THICKNESS
CLT - CLEAT
UNO - UNLESS NOTED OTHERWISE
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
NOZ - NOZZLE
DN - DIAMETER NOMINAL
SCH - SCHEDULE
ELVN - ELEVATION
PROJ - PROJECTION
MM - MILLIMETER
TYP - TYPICAL
GA - GENERAL ARRANGEMENT
DRG - DRAWING
CL - CENTER LINE
TOS - TOP OF STEEL
WP - WORKING POINT
N/S - NEAR SIDE
F/S - FAR SIDE
DIA - DIAMETER
C/F - CUT FORM
TBC - TO BE CONFIRMED
TOG - TOP OF GRATING
N/A - NOT APPLICABLE
ATM - ATMOSPHERIC
MIN - MINIMUM
TRT - TREATMENT
DIST - DISTRIBUTION

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. BOLT HOLES IN NOZZLE FLANGES & MANHOLES SHALL STRADDLE VERTICAL CENTER LINES ON SHELL & NORTH SOUTH CENTERLINES ON ROOF.
5. NOZZLE PROJECTION ON SHELL SIDE REFERS FROM EXTERNAL SURFACE OF SHELL PLATE TO FACE OF FLANGE. NOZZLE PROJECTION ON ROOF SIDE REFERS FROM TOP OF ROOF PLATE TO FACE OF FLANGE.
6. VERTICAL SHELL JOINTS SHOULD PREFERABLY BE OFFSET BY 1000 MM.
7. CURB ANGLE JOINTS SHALL STAGGER ATLEAST 300 MM WITH VERTICAL SHELL PLATE WELD.
8. REINFORCEMENT PADS TO HAVE TELLTALE HOLES OF 6 MM DIA AND SHALL BE TESTED AT 100 kPa PRESSURE WITH SOAP SOLUTION AND THEN FILLED WITH HARD GREASE.
9. FLANGE (COD & MANHOLE) GASKET SEATING SURFACE SHALL BE PROVIDED WITH 0.0032 TO 0.0063 mm (3.2 TO 6.3 µm) ROUGHNESS FINISH.
10. TWO NUMBERS EARTHING LUGS SHALL BE CONSIDERED.
11. COD GROOVE DIMENSION TO BE FINALISED BASED ON COD BOTTOM PLATE AND FOUNDATION RING WIDTH.
12. MINIMUM THICKNESS OF RF PAD, FLANGE AND COVER PLATES FOR NOZZLES ARE PROVIDED BASED ON HYDROSTATIC LOAD AND FINAL SIZE SHALL BE CONFIRMED AFTER NOZZLE DESIGN IN SEPERATE DOCUMENT.
13. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 49WF-EG - ME- GAD - 2006 & 2007.
14. FOR DESIGN OF TANK REFER DOCUMENT NUMBER MD-216-49WF-EG-ME-CAL-2001 AND FOR DESIGN OF TANK ROOF SYSTEM REFER DOCUMENT NUMBER MD-216-3750-EG-ME-CAL-2001.
15. ALL LOADS ARE "UNFACTORED". REFER DOCUMENT NUMBER MD-216-49WF-EG-ME-CAL-2001 FOR ALL LOAD VALUES EXCEPT THE "TANK LIQUID LOAD" VALUES WHICH ARE CALCULATED BASED ON ACTUAL LIQUID VOLUME.
16. DELETED.
17. PAINTING SHALL CONFORM TO MAADEN SPECIFICATION NUMBER MD-101-SMEM-EG-ME-SPC-0007 AND SEPARATE PAINTING DETAILS & APPLICATION PROCEDURE WILL BE SUBMITTED FOR APPROVAL.
18. ALL NOZZLE FLANGES SHALL BE IN ACCORDANCE WITH ASME B16.5 FOR SIZE UPTO DN600 AND TO ASME B16.47 SERIES A FOR DN750 AND ABOVE.
19. FOR HANDRAIL REFER DRG. NO. MD-216-1004-EG-ST-DED-1003 (LATEST).
20. SELF CLOSING HANDRAIL DOOR TO BE CONSIDERED AT TOP OF EACH LADDER.
21. FOR PIPE DN50 AND BELOW SUITABLE PIPE SUPPORT TO BE PROVIDED AT SITE.

DESIGN DATA			
TANK DETAILS	LOCATION		POTABLE WATER TREATMENT AREA
	TANK NAME		RAW WATER STORAGE TANK
	EQUIPMENT TAG No.		49WF-TNK-003
	TANK DIMENSIONS		INSIDE DIAMETER : 5100mm, HEIGHT: 7500mm
	QUANTITY (NUMBER)		1
	ROOF TYPE & SLOPE		SLOPED CONE ROOF (SLOPE 1 IN 12) CONICAL APEX UP
	BOTTOM TYPE & SLOPE		SLOPED BOTTOM (SLOPE 1 IN 120) CONICAL APEX UP
	FIELD ERECTED/SHOP FABRICATED		FIELD ERECTED
	STORED FLUID		WATER DRAWN FROM BOREWELL
DESIGN CODE	DESIGN CODE		API 650-TWELFTH EDITION, MARCH 2013, ADDENDUM 3, AUGUST 2018
OPERATING DATA	OPERATING TEMPERATURE		40° C
	NET WORKING CAPACITY		121 m^3
	MAXIMUM CAPACITY		137 m^3
	OPERATING LEVEL	MINIMUM	420 mm
		MAXIMUM	6690 mm
	OPERATING DENSITY OF CONTENT		1.0 t/m^3
MECHANICAL & PROCESS DATA	DESIGN TEMPERATURE		50° C
	DESIGN PRESSURE (INTERNAL/EXTERNAL)		ATM/ATM
	TEST PRESSURE		ATM
	HYDRO STATIC TEST PRESSURE		FULL OF WATER
	DESIGN DENSITY OF CONTENT		1.0 t/m^3
	TESTING		AS PER API 650
	INSPECTION		AS PER API 650
	DENSITY OF SCALE		N/A
	DESIGN SCALE THICKNESS	TANK SHELL/ROOF	N/A
	DESIGN FILL RATE		20 m/hr
	DESIGN EMPTY RATE		10 m/hr
	CORROSION ALLOWANCE	SHELL	1.6 mm
		ROOF PLATE	0 mm
		BOTTOM PLATE / ANNULAR PLATE	1.6 mm
		ANCHOR BOLT	0 mm
		STRUCTURAL	1.6 mm
WIND LOAD DATA	BASIC WIND SPEED		43 m/sec. (155 Km/hr.)
	SITE EXPOSURE		C
	VELOCITY PRESSURE EXPOSURE COEFFICIENT		K_z= AS PER ELEVATION
	TOPOGRAPHIC FACTOR		K_{zt}=1.0
	WIND DIRECTIONALITY FACTOR		K_d=0.95
SEISMIC LOAD DATA	SITE CLASS		B
	SEISMIC USER GROUP		SUG - I
	IMPORTANCE FACTOR	(I)	1
	SCALING FACTOR	(Q)	2/3
	MAXIMUM DESIGN LIQUID LEVEL	(H)	6.700 m
	CO-EFFICIENT TO ADJUST THE SPECTRAL ACCELERATION FROM 5% TO 0.5% DAMPING (K)		1.50
	SEISMIC CO-EFFICIENTS		S_s= 0.1g, S_1= 0.02g, F_a = 1.0, F_v= 1.0 S_{ds} = 0.067g. S_{d1}= 0.013g, T= 4 sec.
OTHER DATA	ROOF LIFE LOAD		1.2 kPa
	FOUNDATION TYPE (RING WALL/SLAB)		RING WALL
	PWHT REQUIRED(YES/NO)- IF YES,SPECIFY REASON FOR PWHT		NO
	REASON FOR PWHT(PROCESS REASON/PER API CODE)		AS PER API 650
	RADIOGRAPHY		AS PER CL 8.1 - API 650
	AGITATOR RAKE REQUIRED	(YES/NO)	NO
		EQUIPMENT TYPE (AGITATOR/RAKE)	N/A
		AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)	N/A
	INSULATION	SHELL	N/A
		ROOF	N/A
	LINER REQUIRED	(YES/NO)	NO
		LINER MATERIAL	N/A
		LINER THICKNESS (ROOF/SHELL/BOT)	N/A
	PAINTING	INTERNAL	YES (REFER NOTE 17)
		EXTERNAL	YES
	PIPE SUPPORT		YES
	CAGE LADDER		YES
	ROOF CIRCUMFERENTIAL HANDRAIL		YES
	WIND GIRDER		AS REQUIRED
	CATHODIC PROTECTION		NO
	LEAK DETECTION (APPURTENANCE)		NO
	INTERNAL BAFFLES		NO
	EARTHING LUGS		YES

THIS DRAWING SUPERSEDES MAADEN DRAWING NUMBER MD-216-49WF-EG-ST-GAD-1005-C01 DATE: 18/03/2020

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	30.10.2023	AS BUILT	AKDS	PRI	KDU	AG	
		C02	25/07/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED ⟨C02⟩	AKDS	PRI	KDU	AG	
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	C01	18/03/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER'S	SJH	PRI	KDU	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA			COMMENTS ON 12/03/2020 & REVISED AS MKD. C01					
MD-216-49WF-EG-ME-GAD-1007	RAW WATER TANK 49WF-TNK-003	A01	15/02/2020	ISSUED FOR REFERENCE	SJH	PRI	KDU	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER: 019045_080_S_G_26212_001 | REVISION: 3

OUTOTEC DRAWING NUMBER: 900707 __GAU01 __AG32__MTL01 __53063 | DOCUMENT ID: OU1006714.00 | REVISION: 3

DRAWN	SJH	11/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	PRI	14/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	KDU	13/02/2020	POTABLE WATER SUPPLY, TRT & DIST
APPROVED	AG	15/02/2020	PLAN, ELVN, DETAIL OF TANK 49WF-TNK-003
EMPLOYER			
SCALE	1:100,20	A0	DRAWING NUMBER: MD - 216 - 49WF-EG - ME - GAD - 2005 | REV: D01